CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: December 2020 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Find the real root of the equation $x^3 - x - 1 = 0$.**

Solution: $f(x) = x^3 - x - 1$.

Using Newton-Raphson method with initial guess $x_0 = 1.3$: $f'(x) = 3x^2 - 1$.

$f(1.3) = 2.197 - 1.3 - 1 = -0.103$.

$f'(1.3) = 3(1.69) - 1 = 4.07$.

$x_1 = 1.3 - \frac{-0.103}{4.07} = 1.3 + 0.0253 = 1.3253$.

$f(1.3253) \approx 0.002$.

Root: $\mathbf{x \approx 1.325}$.

Q2 (3M) **Define operators E, Δ and ∇. Also obtain relation between E and Δ.**

Definitions:

1. **Shift Operator E:** $E\, f(x) = f(x + h)$.
2. **Forward Difference Δ:** $\Delta f(x) = f(x + h) - f(x)$.
3. **Backward Difference ∇:** $\nabla f(x) = f(x) - f(x - h)$.

Relation between E and Δ:

$\Delta f(x) = f(x + h) - f(x) = E f(x) - f(x) = (E - 1) f(x) \implies \mathbf{\Delta = E - 1 \iff E = 1 + \Delta}$.

Q3 (3M) **Given dy/dx = (y - x)/(y + x) with y(0) = 1. Find y(0.3) by Euler's method taking h = 0.1.**

Solution: $y_{n+1} = y_n + h \frac{y_n - x_n}{y_n + x_n}$, $h = 0.1$.

1. $x_0 = 0, y_0 = 1 \implies y_1 = 1 + 0.1 \left(\frac{1 - 0}{1 + 0}\right) = 1 + 0.1(1) = \mathbf{1.1000}$ at $x = 0.1$.
2. $x_1 = 0.1, y_1 = 1.1 \implies y_2 = 1.1 + 0.1 \left(\frac{1.1 - 0.1}{1.1 + 0.1}\right) = 1.1 + 0.1\left(\frac{1.0}{1.2}\right) = 1.1 + 0.0833 = \mathbf{1.1833}$ at $x = 0.2$.
3. $x_2 = 0.2, y_2 = 1.1833 \implies y_3 = 1.1833 + 0.1 \left(\frac{1.1833 - 0.2}{1.1833 + 0.2}\right) = 1.1833 + 0.1\left(\frac{0.9833}{1.3833}\right) = 1.1833 + 0.0711 = \mathbf{1.2544}$ at $x = 0.3$.

Answer: $\mathbf{y(0.3) \approx 1.254}$.

Q4 (3M) **Define Simpson's three-eighths rule for Numerical Integration.**

Definition: Simpson's 3/8 rule approximates the integral of $f(x)$ over $[a, b]$ divided into n subintervals where n is a multiple of 3 by fitting cubic polynomials across triplets of intervals:

$$\int_a^b f(x)\, dx \approx \frac{3h}{8} \left[(y_0 + y_n) + 3\sum_{i \neq 3k} y_i + 2\sum_{i = 3k} y_i \right]$$

The truncation error is of order $O(h^4)$, specifically $E = -\frac{3}{80} h^5 f^{(4)}(\xi)$.

Q5 (3M) **What do you understand by Gauss's Elimination method?**

Explanation: Gauss Elimination is a direct algorithm for solving linear systems $AX = B$ in two stages:

1. **Forward Elimination:** Performs elementary row operations on the augmented matrix $[A|B]$ to reduce it to an upper triangular form. Partial pivoting (swapping rows for the largest absolute pivot) is used to maintain numerical stability and avoid division by zero.
2. **Back-Substitution:** Solves the unknowns starting from the last row upwards.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Use Picard's method to approximate y when x = 0.2, given dy/dx = x² + y² with y(0) = 0.**

Solution: x_0 = 0, y_0 = 0.
y^{(1)}(x) = 0 + \int_0^x (t^2 + 0) dt = \frac{x^3}{3}.
y^{(2)}(x) = \int_0^x \left(t^2 + \frac{t^6}{9} \right) dt = \frac{x^3}{3} + \frac{x^7}{63}.
At x = 0.2: y(0.2) = \frac{(0.2)^3}{3} + \frac{(0.2)^7}{63} = \frac{0.008}{3} + \frac{0.0000128}{63} = 0.002667 + 0.000000 = \mathbf{0.00267}.

Q7 (7.5M) **Find backward differences for the points: (5, 11), (6, 13), (9, 28), (11, 40).**

Solution: Since step size is unequal, using divided differences: f[5, 6] = \frac{13 - 11}{6 - 5} = 2.
f[6, 9] = \frac{28 - 13}{9 - 6} = \frac{15}{3} = 5.
f[9, 11] = \frac{40 - 28}{11 - 9} = \frac{12}{2} = 6.
Second divided differences: f[5, 6, 9] = \frac{5 - 2}{9 - 5} = \frac{3}{4} = 0.75.
f[6, 9, 11] = \frac{6 - 5}{11 - 6} = \frac{1}{5} = 0.20.

Q8 (7.5M) **Using Bessel's formula, find y(25) from: x = [20, 24, 28, 32], y = [24, 32, 35, 40].**

Solution: h = 4. Choose origin x_0 = 24. Then u = \frac{25 - 24}{4} = 0.25.
y_{-1} = 24, y_0 = 32, y_1 = 35, y_2 = 40.
\Delta y_0 = 35 - 32 = 3, \Delta y_{-1} = 32 - 24 = 8, \Delta y_1 = 40 - 35 = 5.
\Delta^2 y_{-1} = 3 - 8 = -5, \Delta^2 y_0 = 5 - 3 = 2.
Bessel's interpolation formula:

> y = \frac{y_0 + y_1}{2} + \left(u - \frac{1}{2}\right)\Delta y_0 + \frac{u(u-1)}{4}(\Delta^2 y_{-1} + \Delta^2 y_0)

\frac{32 + 35}{2} + (0.25 - 0.5)(3) + \frac{0.25(-0.75)}{4}(-5 + 2)
= 33.5 - 0.75 + \frac{-0.1875}{4}(-3) = 32.75 + 0.1406 = \mathbf{32.89}.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Apply Gauss-Seidel iteration method to solve: 20x + y - 2z = 17, 3x + 20y - z = -18, 2x - 3y + 20z = 25.**

Solution: Iteration equations:

> x = \frac{17 - y + 2z}{20}, \quad y = \frac{-18 - 3x + z}{20}, \quad z = \frac{25 - 2x + 3y}{20}

Iter 1: x_1 = 0.8500, y_1 = -1.0275, z_1 = 1.0109.
Iter 2: x_2 = 1.0025, y_2 = -0.9998, z_2 = 0.9998.
Iter 3: x_3 = 1.0000, y_3 = -1.0000, z_3 = 1.0000.
Answer: \mathbf{x = 1.000, \; y = -1.000, \; z = 1.000}.

Q10 (15M) **Use Runge-Kutta method of fourth order to find y(0.2) for dy/dx = y + x with y(0) = 1 (take h = 0.2).**

Solution: $f(x, y) = y + x$, $x_0 = 0, y_0 = 1, h = 0.2$.

$k_1 = 0.2(1 + 0) = 0.2000$

$k_2 = 0.2(1.1 + 0.1) = 0.2400$

$k_3 = 0.2(1.12 + 0.1) = 0.2440$

$k_4 = 0.2(1.244 + 0.2) = 0.2888$

$\Delta y = \frac{0.2000 + 2(0.2400) + 2(0.2440) + 0.2888}{6} = 0.2428$

> $\mathbf{y(0.2) = 1.2428}$

Q11 (15M) **Evaluate $\int_0^6 1/(1+x^2)\,dx$ by using: (a) Trapezoidal rule, (b) Simpson's 3/8 rule.**

Solution: $h = 1$, $y = [1.0000, 0.5000, 0.2000, 0.1000, 0.0588, 0.0385, 0.0270]$.

(a) Trapezoidal Rule:

> $I_T = \frac{h}{2} [(y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5)] = 0.5 \times [1.0270 + 2(0.8973)] = 0.5 \times 2.8216 = \mathbf{1.4108}$

(b) Simpson's 3/8 Rule:

> $I_{3/8} = \frac{3}{8} [(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)] = \frac{3}{8} [1.0270 + 3(0.7973) + 0.2000] = \mathbf{1.3571}$

Q12 (15M) **By using Newton-Raphson's method, find the root of $x^3 - 2x - 5 = 0$ near x = 2 correct to 3 decimal places. Also obtain the rate of convergence of Newton-Raphson method.**

Solution: $f(x) = x^3 - 2x - 5$, $f'(x) = 3x^2 - 2$.

Starting at $x_0 = 2$:

$f(2) = -1$, $f'(2) = 10 \implies x_1 = 2 - \frac{-1}{10} = 2.1000$.

$f(2.1) = 9.261 - 4.2 - 5 = 0.061$, $f'(2.1) = 3(4.41) - 2 = 11.23$.

$x_2 = 2.1 - \frac{0.061}{11.23} = 2.1 - 0.00543 = 2.09457 \implies \mathbf{x \approx 2.095}$.

Rate of Convergence Proof:

Let α be the exact root, $x_n = \alpha + \epsilon_n$.

By Taylor expansion: $f(x_n) = f(\alpha) + \epsilon_n f'(\alpha) + \frac{\epsilon_n^2}{2} f''(\alpha) + \dots = \epsilon_n f'(\alpha) + \frac{\epsilon_n^2}{2} f''(\alpha)$.

$f'(x_n) = f'(\alpha) + \epsilon_n f''(\alpha)$.

Substitute into NR formula: $\epsilon_{n+1} = \epsilon_n - \frac{\epsilon_n f'(\alpha) + \frac{\epsilon_n^2}{2}f''(\alpha)}{f'(\alpha) + \epsilon_n f''(\alpha)} = \epsilon_n - \epsilon_n \left(1 + \frac{\epsilon_n f''(\alpha)}{2 f'(\alpha)}\right)\left(1 - \frac{\epsilon_n f''(\alpha)}{f'(\alpha)}\right) \approx \frac{f''(\alpha)}{2 f'(\alpha)} \epsilon_n^2$.

Since $\epsilon_{n+1} \propto \epsilon_n^2$, the convergence is **quadratic (order = 2)**.

Q13 (15M) **Apply Lagrange's formula to find the cubic polynomial which includes: (0, 2), (1, 3), (2, 12), (5, 147).**

Solution: Applying Lagrange interpolation across $x = [0, 1, 2, 5]$, $y = [2, 3, 12, 147]$:

Notice $y = x^3 + 2$:

$x=0 \implies y=2;$

$x=1 \implies y=3;$

$x=2 \implies y=8+2=10 \approx 12.$

Computing Lagrange polynomial:

$$\mathbf{y(x) = x^3 + x^2 - x + 2}$$

Verification: $y(0)=2, y(1)=3, y(2)=8+4-2+2=12, y(5)=125+25-5+2=147.$ (Exact!)